



Intro to Local LLMs

Kei Tang

Recurse Center

2026-09-23



Outline

1. Introduction
2. Model sizes and memory: What model can I run?
3. GPUs and optimizations: How fast will the model run?
4. Practical stuff
5. Demo



Outline

- 1. Introduction**
2. Model sizes and memory: What model can I run?
3. GPUs and optimizations: How fast will the model run?
4. Practical stuff
5. Demo



1.1 Why local LLMs?

- Privacy: Cloud providers may retain your data or use it for training
- Cost: Powerful cloud models are expensive
- Research and learning: Know more about how LLMs work
- and more?



1.2 What I will not cover

- Harness: Models are generally decoupled from harnesses



1.2 What I will not cover

- Harness: Models are generally decoupled from harnesses
- Training: Maybe another workshop in the future?



1.2 What I will not cover

- Harness: Models are generally decoupled from harnesses
- Training: Maybe another workshop in the future?
- Platform specific things (e.g. Metal for Macs, Nvidia stuff): I don't own those hardware, but we can look into it together!



Outline

1. Introduction
- 2. Model sizes and memory: What model can I run?**
3. GPUs and optimizations: How fast will the model run?
4. Practical stuff
5. Demo



2.1 Getting a feel on different model sizes



2.2 vRAM, RAM and unified memory



2.3 Quantization



2.4 Using disk space? (Advanced)



Outline

1. Introduction
2. Model sizes and memory: What model can I run?
- 3. GPUs and optimizations: How fast will the model run?**
4. Practical stuff
5. Demo



3.1 How “speed” is measured



3.2 Factors affecting “speed”



3.3 Optimizations: MTP



3.4 Optimizations: NVFP4 (Advanced)



Outline

1. Introduction
2. Model sizes and memory: What model can I run?
3. GPUs and optimizations: How fast will the model run?
- 4. Practical stuff**
5. Demo



4.1 Where to find & download models?



4.2 All the different frameworks!



Outline

1. Introduction
2. Model sizes and memory: What model can I run?
3. GPUs and optimizations: How fast will the model run?
4. Practical stuff
- 5. Demo**



5.1 Setting up a local LLM, live

1. Choosing a (quantized) model
2. Running it in llama.cpp
3. Running it in Ramalama